

[This question paper contains 8 printed pages.]

Your Roll No.....

Sr. No. of Question Paper : 1231

C

Unique Paper Code : 32357505

Name of the Paper : DSE-2 Discrete Mathematics

Name of the Course : B.Sc. (H) Mathematics

Semester : V (under CBCS (LOCF)
Scheme)

Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. All the given **eight** questions are compulsory to attempt.
3. Do any **two** parts from each of the given **eight** questions.
4. Marks for each part are indicated on the right in brackets.

P.T.O.

SECTION I

1. (a) Let N_0 be the set of non-negative integers. Define a relation $\leq$ on N_0 as: For $m, n \in N_0$, $m \leq n$ if m divides n , that is, if there exists $k \in N_0$: $n = km$. Then show that $\leq$ is an order relation on N_0 . (2½)
- (b) If '1', '2', '3' denote chains of one, two, three elements respectively and $\bar{3}$ denotes anti chain of three elements, then draw the Hasse diagram for the dual of $L \oplus K$ when $L = \bar{3}$ and $K = 1 \oplus (2 \times 2)$. (2½)
- (c) Define maximum and a maximal element of a partially ordered set P . Give an example each for both definitions. (2½)
2. (a) Let P and Q be finite ordered sets and let $\psi: P \rightarrow Q$ be a bijective map. Then show that the following are equivalent :
- (i) $x < y$ in P iff $\psi(x) < \psi(y)$ in Q
 - (ii) $x \prec y$ in P iff $\psi(x) \prec \psi(y)$ in Q (3)

(b) Define upper bound and lower bound of a subset S of a partially ordered set P . Construct an example of a partially ordered set P and its subset S and give the set of all upper bounds and lower bounds of S . (3)

(c) Let P and Q be ordered sets. Then show that the ordered sets P and Q are order isomorphic iff there exist order preserving maps $\phi: P \rightarrow Q$ and $\psi: Q \rightarrow P$ such that :

$$\phi \circ \psi = \text{id}_Q \text{ and } \psi \circ \phi = \text{id}_P \text{ where } \text{id}_S: S \rightarrow S \text{ denotes the identity map on } S \text{ given by: } \text{id}_S(x) = x, \forall x \in S. \quad (3)$$

SECTION II

3. (a) Let $D_{60} = \{1, 2, 4, 5, 6, 12, 20, 30, 60\}$ be an ordered subset of $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$, $\mathbb{N}$ being the set of natural numbers. If ' $\leq$ ' is defined on D_{60} by $m \leq n$ if and only if m divides n then show that D_{60} does not form a lattice. Also Draw the diagram of D_{60} and find elements $a, b, c, d \in D_{60}$ such that $a \vee b$ and $c \wedge d$ do not exist in D_{60} . (5½)

(b) Define sublattice of a lattice. Prove that every chain of a lattice L is a lattice and also a sublattice of L . (5½)

P.T.O.

(c) Define modular lattice. Prove that a homomorphic image of modular lattice is modular. $(5\frac{1}{2})$

4. (a) Let L be a lattice. For any $a, b, c \in L$, show that the following inequalities hold :

$$(i) a \wedge (b \vee c) \geq (a \wedge b) \vee (a \wedge c)$$

$$(ii) a \geq c \Rightarrow a \wedge (b \vee c) \geq (a \wedge b) \vee c \quad (5)$$

(b) Let $(L, \wedge, \vee)$ be an algebraic lattice. If we define

$$a \leq b : \Leftrightarrow a \vee b = b$$

then show that $(L, \leq)$ is a lattice ordered set. (5)

(c) Let L_1 and L_2 be distributive lattices. Prove that the product $L_1 \times L_2$ is a distributive lattice. (5)

SECTION III

5. (a) A voting-machine for three voters has YES-NO switches. Current is in the circuit precisely when YES has a majority. Draw the corresponding contact diagram and the switching/circuit diagram. $(5\frac{1}{2})$

(b) Show that a Boolean Algebra is relatively complemented. (5½)

(c) Simplify the polynomial :

$$f = x'yz + x'yz' + x'y'z + xy'z' + xy'z$$

using Quine's McCluskey method. (5½)

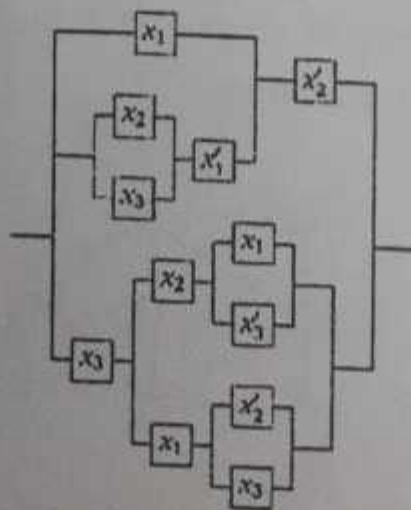
6. (a) Define a system of normal forms. Find conjunctive normal form for $p = y'z' + x'yz$. (5)

(b) Simplify the Boolean expression :

$$f = w'xy'z + w'xyz + w'xyz' + wxy'z + wxyz + wxyz' + wx'y'z + wx'yz$$

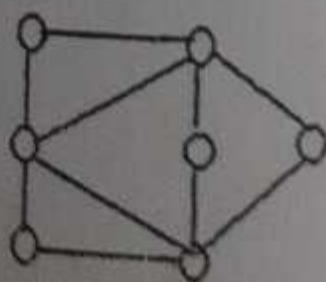
using Karnaugh Diagram. (5)

(c) Find the symbolic gate representation of the contact diagram : (5)

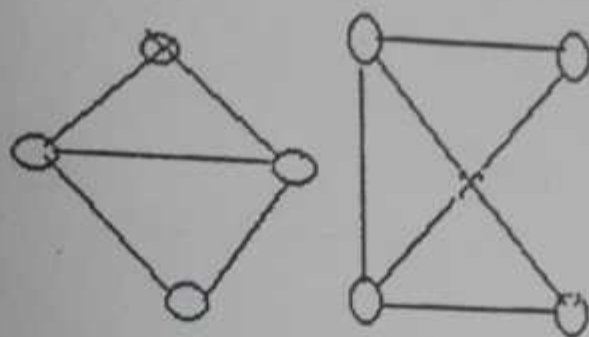


SECTION IV

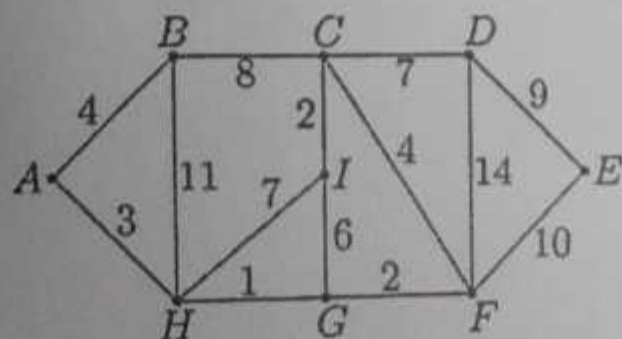
7. (a) (i) Show that the sum of the degrees of the vertices of a pseudograph is an even number equal to twice the number of edges.
- (ii) A graph has five vertices of degree 4 and two vertices of degree 2. How many edges does it have? $(5\frac{1}{2})$
- (b) (i) Define the degree sequence of a graph. Does there exist a graph with following degree sequence 6, 6, 5, 5, 4, 4, 4, 4, 3?
- (ii) Show that the number of vertices of odd in a graph must be even. $(5\frac{1}{2})$
- (c) (i) What is a bipartite graph? Determine whether the graph given below is bipartite or not. Give the bipartition sets or explain why the graph is not bipartite.



- (ii) Define isomorphism of graphs. Also label the following graphs so as to show an isomorphism. (5½)

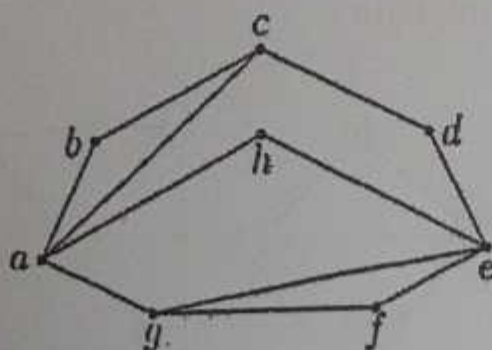


8. (a) Construct a Gray Code of length 3 using the concept of Hamiltonian Cycles. (5½)
- (b) Apply Dijkstra's algorithm to find a shortest path from A to all other vertices in the weighted graph shown. (5½)



- (c) (i) Does there exist a graph G with 28 edges and 12 vertices each of degree 3 or 6?

- (ii) Define Eulerian circuit. Is the given graph Eulerian? Give reasons for your answer.



(5½)